

CS5330 Final Project 25Spring

- AI Fitness Trainer - Real-Time Pose Correction and Feedback

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Problem Statement

This project aims to build an AI-powered fitness trainer that provides real-time feedback on workout form using computer vision. The system takes a live or recorded video of a user performing exercises (such as squats, push-ups, or planks) as input and outputs pose evaluations, form correction feedback, and repetition counts. Instead of relying on human trainers, users will receive automatic voice and visual guidance to ensure proper exercise technique, reducing injury risk and improving effectiveness.

How to Obtain Data/Model Code

To avoid collecting and annotating a large dataset, we will use MediaPipe Pose, a pre-trained deep learning model developed by Google that detects 33 key body landmarks in real time. MediaPipe provides accessible Python APIs and requires no training on our end. For initial testing, we will use public YouTube fitness videos and webcam-recorded clips of ourselves performing exercises to validate and fine-tune the angle thresholds for feedback logic.

Design of Our Solution

Our system uses MediaPipe to extract keypoint coordinates of the user's body in real time. We store and track the joint positions frame-by-frame to enable dynamic tracing and motion analysis. For each supported exercise (e.g., squats, push-ups, planks), we define custom logic based on joint angles and body part positions to detect common form mistakes, such as "knees caving in" or "hips too high." Using this logic, we classify the correctness of the current pose and provide immediate feedback through on-screen text and voice alerts. Additionally, we implement automatic exercise type classification, form scoring per rep, and real-time repetition counting without user setup. Each exercise type uses a different pose evaluation logic, as shown in the attached reference table.

Computing Power and Training Needs

Since MediaPipe is pre-trained and optimized for real-time inference, we do not need to train any deep learning models. All computations will be performed locally using personal laptops. The MediaPipe runtime is lightweight and runs efficiently on CPU, making it feasible for live video analysis without external GPU resources.

Expected Outcome

A successful outcome is a real-time feedback system that supports at least three types of exercises (squats, push-ups, planks), identifies and tracks common posture errors based on biomechanical rules, provides audio-visual cues when mistakes occur, counts proper repetitions, and assigns a form score per rep (and per set). The system should not require prior setup by the user and should adaptively recognize the performed exercise type.

Reference

https://ai.google.dev/edge/mediapipe/solutions/vision/pose_landmarker/python

Squats:

Mistake	Detection Logic	Feedback
Not deep enough	Knee angle > 90°	"Go lower!"
Knees caving in	Distance between knees < threshold	"Keep knees aligned!"
Leaning too forward	Chest to hip angle < threshold	"Keep back straight!"

Push-ups:

Mistake	Detection Logic	Feedback
Hips too high	Hip higher than shoulders & ankles	"Lower your hips!"
Arms flaring	Elbow angle > 90°	"Keep elbows in!"
Not going low enough	Elbow < 90°	"Go deeper!"

Planks:

Mistake	Detection Logic	Feedback
Hips sagging	Hip lower than shoulders	"Raise your hips!"
Head position bad	Head tilt angle off from spine	"Keep head neutral!"